

AMENDMENT NO. 1 AUGUST 2023
TO
IS 12084 : 2018 MORPHOLINE — SPECIFICATION
(*First Revision*)

(Foreword, para 2, line 2) — Delete.

(Page 1, clause 1) — Substitute the following for the existing:

‘This standard prescribes the requirements and methods of sampling and test for morpholine.’

(Page 1, Table 1) — Substitute the following for the existing Table 1:

SI No.	Characteristic	Requirements	Method of Test, Ref to	
			Annex (4)	IS No. (5)
(1)	(2)	(3)	(4)	(5)
i)	Purity (as morpholine), percent by mass, <i>Min</i>	99.0	B	—
ii)	Relative density, 20/20 °C, <i>Min</i>	1.001	C	—
iii)	(Pt-Co) colour scale, <i>Max</i>	15	—	3025 (Part 4)
iv)	Boiling range	125 °C to 129 °C for 90 percent recovery	—	5298
v)	Residue, on evaporation, mg/l, <i>Max</i>	1 000	D	—
vi)	Ash content, g/100 ml, <i>Max</i>	0.003	E	—
vii)	Iron (as Fe), ppm, <i>Max</i>	5	F	3025 (Part 53) or 3025 (Part 2) ¹⁾ or 3025 (Part 65) ¹⁾
viii)	Copper (as Cu), ppm, <i>Max</i>	5	F	3025 (Part 42) or 3025 (Part 2) ¹⁾ or 3025 (Part 65) ¹⁾
ix)	Nickel (as Ni), ppm, <i>Max</i>	5	F	3025 (Part 54) or 3025 (Part 2) ¹⁾ or 3025 (Part 65) ¹⁾
x)	Silica (as SiO ₂), ppm, <i>Max</i>	5	—	3025 (Part 35) or 3025 (Part 2) ²⁾ or 3025 (Part 65) ²⁾
xi)	Chloride (as Cl), ppm, <i>Max</i>	5	—	3025 (Part 32)

¹⁾ In case of determination of Cu, Fe, and Ni in morpholine by IS 3025 (Part 2) or IS 3025 (Part 65), it is recommended that the sample preparation as prescribed in **F-1** of Annex F may be followed.

²⁾ In case of determination of Silica (as SiO₂) in morpholine by IS 3025 (Part 2) or IS 3025 (Part 65), it is recommended that the sample preparation as prescribed in **F-2** of Annex F may be followed.

Amendment No. 1 to IS 12084 : 2018

(Page 3, Annex A) — Insert the following after IS 2263 : 1979:

<i>IS No.</i>	<i>Title</i>
IS 3025 (Part 2) : 2019/ISO 11885 : 2007	Methods of sampling and test (physical and chemical) for water and wastewater: Part 2 Determination of selected elements by inductively coupled plasma optical emission spectrometry (ICP - OES) (<i>first revision</i>)
IS 3025 (Part 65) : 2022/ISO 17294-2 : 2016	Methods of sampling and test (physical and chemical) for water and wastewater: Part 65 Application of inductively coupled plasma mass spectrometry (ICP - MS) — Determination of selected elements including uranium isotopes (<i>first revision</i>)

(Page 5, Annex E) — Insert the following Annex after Annex E:

‘ANNEX F

[Table 1, *Sl No.* (vii), (viii), (ix), and (x)]

**SAMPLE PREPARATION FOR DETRMINATION OF COPPER, IRON, NICKLE, AND SILICA IN
MORPHOLINE BY ICP-OES OR ICP-MS METHOD**

**F-1 SAMPLE PREPARATION METHOD FOR DETERMINATION OF COPPER, IRON AND NICKLE
IN MORPHOLINE USING ICP OES OR ICP-MS**

F-1.1 Automated Digestion System

F-1.1.1 Analytes — Cu, Fe, Ni

F-1.1.2 Reagents

F-1.1.2.1 De-ionized water

F-1.1.2.2 Sulphuric acid (H₂SO₄), 96 percent (m/m)

F-1.1.2.3 Perchloric acid (HClO₄), 70 percent (m/m)

F-1.1.2.4 Nitric acid (HNO₃), 65 percent (m/m)

F-1.1.2.5 Hydrochloric acid (HCl), 36 perecent (m/m)

F-1.1.2.6 Cesium sulphate (Cs₂SO₄), 99.9 percent (m/m)

F-1.1.2.7 Mixed acid

Mix concentrated nitric acid, concentrated sulphuric acid and concentrated perchloric acid at a volume ratio of 2 : 1 : 1.

F-1.1.2.8 Acid mixture 1

Mix sulfuric acid and nitric acid at a volume ratio of 39 : 1. Add 2.2 g/l Cs₂SO₄ to it.

F-1.1.2.9 Hydrochloric acid, 5 percent (v/v)

Mix hydrochloric acid 36 percent (m/m) and de-ionized water in a volume ratio of 1 : 19.

F-1.3 Sample Preparation

Weigh 0.35 g to 0.40 g of sample and add it into the automated acid digestion system. In digestion system, add acid mixture-I and crack the sample at 320 °C approximately. Digest completely the organic remnants with mixed acid at 160 °C. Evaporate the excess acids to dryness. Add 5 percent (v/v) hydrochloric acid to the residue, heat until boiling. Perform the analysis on duplicate. Run a blank in an analogous manner.

F-2 SAMPLE PREPARATION METHOD FOR DETERMINATION OF SILICA IN MORPHOLINE USING ICP OES OR ICP-MS

F-2.1 Reagents

F-2.1.1 *Hydrochloric Acid* (HCl), 50 volume percent, 6 ml

F-2.1.2 *Potassium Carbonate-Sodium Carbonate* (K_2CO_3 - Na_2CO_3), 0.4 g

F-2.1.3 *Borax* ($Na_2B_4O_7$), 0.1 g

F-2.2 Apparatus

F-2.2.1 *Dessicator*

F-2.2.2 *Muffle Furnace*, calibrated to 550 °C

F-2.2.3 *Platinum Dish*

F-2.2.4 *Balance*, capable of weighing the density bottle to the nearest 0.1 mg

F-2.2.5 *Open Flame (Bunsen Burner)*

F-2.3 Sample Preparation

F-2.3.1 Pre-condition the Pt dish in muffle furnace at 550 °C for 10 min. Cool it down to room temperature in dessicator prior to use. Measure the tare weight of the Pt dish.

Take 100 g to 115 g of sample in small portions of 20 g to 40 g into the Pt dish and burn above the open flame (Bunsen burner) in subsequent steps prior to total incineration. Incinerate the overall sample residue in a muffle furnace at 550 °C for 30 min. Transfer the dish to a dessicator and cool down the residue to room temperature. Carry out the same procedure on duplicate.

F-2.3.2 After incineration, the residue is homogenized with a mixture of K_2CO_3 - Na_2CO_3 and $Na_2B_4O_7$ [soda : borax, 4 : 1 (w/w)] and subsequently melt above an open flame. At the end of the procedure a clear melt is obtained. Dissolve the cooled melt cake in 6 ml of hydrochloric acid and 10 ml of water and heat it. Add water to make up volume of 50 ml.

F-2.3.3 Carry out the digestion and determination on duplicate. The reported value is the mean value of the double determination.'